

Suhas Dasenahalli Lingaraju

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Google Scholar Profile

Summary

My research focuses on extreme events and their compounding, where multiple hazards interact to produce an outcome more severe than any single hazard would on its own. By integrating theory, observations, and modeling, I seek to understand their mechanisms and translate these insights into improved forecasting and early-warning systems.

Research Experience

Postdoctoral Research Associate

Atmospheric and Oceanic Sciences,
University of Colorado, Boulder

2024–present
Advisor: Prof. Weiqing Han

- Demonstrated how atmospheric oscillations drive marine heatwaves in the Arabian Sea, producing a compound ocean-atmosphere extreme that in turn drives extreme precipitation across South Asia.
- Investigated the drivers of sea level extremes along the U.S. East Coast, finding that remote teleconnections from the Indian Ocean can heighten this risk.
- Assessed how well Earth system models represent U.S. East Coast sea level extremes, showing that internal variability by itself can drive much of the disagreement between models.
- Investigating the global impacts of the Madden-Julian Oscillation on compound ocean heat-height extremes, and how these ocean extremes feed back to influence atmospheric hazards, including extreme precipitation.

Postdoctoral Scholar

Earth and Planetary Science,
University of California, Berkeley

2021–2023
Advisor: Prof. William Boos

- Linked monsoon low pressure systems (LPS) to over half of South Asia’s hydro-meteorological disasters, and, more broadly, to precipitation extremes worldwide when co-occurring with mesoscale convective systems.
- Investigated the growth mechanisms of monsoon LPS and assessed their forecast skill across multiple ensemble prediction systems.
- Led the development of the first automated operational forecasting system for monsoon LPS, now deployed in real time with national meteorological agency in India to support early warning and reduce societal risk.

Education

PhD, Atmospheric Science

2021

Indian Institute of Science, Bangalore

Advisor: Prof. Jai Sukhatme

Thesis: “Moist and dry dynamics in shallow water and general circulation models”

- Investigated the influence of water vapor on atmospheric dynamics using a hierarchy of models, from an idealized moist shallow water model to complex general circulation models.
- Identified a distinct eastward propagating “moist Rossby” wave characterized by moist potential vorticity conservation.
- Showed how moisture governs the width and strength of the Hadley cell in an aquaplanet general circulation model, producing intraseasonal variability resembling the MJO.
- Examined the role of moisture in reshaping the turbulence and energy spectrum of atmospheric flow, altering the scale at which moist convection organizes itself.

Master of Technology, Climate Science

2014

Indian Institute of Science, Bangalore

Bachelor of Engineering, Electrical Engineering

2012

Sri Jayachamarajendra College of Engineering, Mysore

Research Funding

- “Roles of climate modes and inter-basin interaction in regulating upper-ocean heat redistribution and extreme sea level events in the western tropical Pacific and southeast Indian Ocean.” NASA Ocean Surface Topography Science Team (OSTST), Awarded 2026.
PI: Weiqing Han; Co-I: **Suhas, D. L.**
- “Global impacts of the Madden-Julian Oscillation on marine heatwaves, sea surface height extremes, and compound heat-height extremes, and their feedback on extreme precipitation.” NSF Atmospheric and Geospace Sciences (AGS), Under Review.
PI: **Suhas, D. L.**; Co-PI: Weiqing Han.
- “Influence of Marine Heatwaves on Atmospheric Rivers and Extreme Precipitation over the Central and Eastern United States.”
NASA ROSES, Sub-Element 5 (Cross-Cutting Research: Inter-Sphere Interactions), Notice of Intent Submitted.
PI: **Suhas, D. L.**; Co-PI: Weiqing Han.

Publications

- **Suhas, D. L.**, Han, W., Shinoda, T., Sun, R., Subramanian, A., Bourassa, M., and Alexander, M. “Marine Heatwaves in the Arabian Sea: Drivers and Impacts on Atmospheric Circulation and Extreme Precipitation.” *Journal of Climate*, e250458. <https://doi.org/10.1175/JCLI-D-25-0458.1>
- Zhu, Y., Han, W., Garuba, O., **Suhas, D. L.**, Leung, L. Y. R., Subramanian, A. C., Balaguru, K., Chen, G., Van Roekel, L., and Wang, J. (2026). Large-scale climate oscillations induce accelerated high tide flooding along the US East Coast. *Communications Earth & Environment*. <https://doi.org/10.1038/s43247-026-03932-y>

- Tsai, W., Duan, S., O'Brien, T. A., Catto, J. L., Ullrich, P. A., Zhou, Y., Leung, L. R., Feng, Zhe., Boos, W. R., **Suhas, D. L.**, Ahmed, F., and Neelin, J. D. (2025). Co-occurring atmospheric features and their contributions to precipitation extremes. *Journal of Geophysical Research: Atmospheres*, 130(5), e2024JD041687. <https://doi.org/10.1029/2024JD041687>
- **Suhas, D. L.**, Vishnu, S., Goyal, S., Sarkar, S., Mukhopadhyay, P., Ullrich, P. A., and Boos, W. R. (2024). Automated operational forecasting of monsoon low pressure systems. *Bulletin of the American Meteorological Society*, 105(12), E2444-E2460. <https://doi.org/10.1175/BAMS-D-23-0067.1>
- **Suhas, D. L.**, and Boos, W. R. (2024). "Evaluating ensemble predictions of South Asian monsoon low pressure system genesis." *Weather and Forecasting*, 39(10), 1377-1386. <https://doi.org/10.1175/WAF-D-24-0044.1>
- **Suhas, D. L.**, Ramesh, N., Kripa, R. M., and Boos, W. R. (2023). "Influence of monsoon low pressure systems on South Asian disasters and implications for disaster prediction." *npj Climate and Atmospheric Science* 6(1), 48. <https://doi.org/10.1038/s41612-023-00376-5>
- **Suhas, D. L.** and Boos, W. R. (2023). "Monsoon depression amplification by horizontal shear and humidity gradients: A shallow water perspective." *Journal of the Atmospheric Sciences*, 80(2), 633-647. <https://doi.org/10.1175/JAS-D-22-0146.1>
- **Suhas, D. L.**, Sukhatme, J., and Harnik, N. (2022). "Dry and moist atmospheric circulation with uniform sea-surface temperature." *Quarterly Journal of the Royal Meteorological Society*, 148(742), 35-56. <https://doi.org/10.1002/qj.4191>
- Schrötte, J., **Suhas, D. L.**, Harnik, N. and Sukhatme, J. (2021), "Turbulence and equatorial waves in moist and dry shallow-water flow, excited through mesoscale stochastic forcing." *Quarterly Journal of the Royal Meteorological Society*, 148(743), 599-619. <https://doi.org/10.1002/qj.4220>
- **Suhas, D. L.**, and Sukhatme, J. (2020). "Moist shallow-water response to tropical forcing: Initial-value problems." *Quarterly Journal of the Royal Meteorological Society*, 146(733), 3695-3714. <https://doi.org/10.1002/qj.3867>
- **Suhas, D. L.**, Sukhatme, J., and Monteiro, J. M. (2017). "Tropical vorticity forcing and superrotation in the spherical shallow-water equations." *Quarterly Journal of the Royal Meteorological Society*, 143(703), 957-965. <https://doi.org/10.1002/qj.2979>
- **Suhas, D. L.**, and Sukhatme, J. (2015). "Low frequency modulation of jets in quasi-geostrophic turbulence." *Physics of Fluids*, 27(1), 016601. <https://doi.org/10.1063/1.4905710>

In review

- **Suhas, D. L.**, Han, W., Zhu, Y., Subramanian, A., Leung, R., Garuba, O., Balaguru, K., and Van Roekel, L. "How Well Do State-of-the-Art Earth System Models Represent the Observed Sea Level Variability and Extremes Along the U.S. East Coast?" *Journal of Advances in Modeling Earth Systems* (In review).
- Zhu, Y., Han, W., Wang, J., **Suhas, D. L.**, Subramanian, A., Leung, R., Garuba, O., Balaguru, K., and Chen, G. "Decadal Sea Level Variability along the U.S. East Coast: Roles of Extratropical versus Tropical Climate Modes over the North Atlantic Ocean." *Geophysical Research Letters* (In review).

Selected Conference Presentations

- **Suhas, D. L.**, Han, W., and others. (2025). “Marine Heatwaves in the Arabian Sea Drive Extreme Precipitation Along the West Coast of India and Across Pakistan.” AGU 2025 annual meeting.
- **Suhas, D. L.** and Boos, W. R. (2023). “The impact of ENSO on tropical cyclone seeds.” AGU Fall Meeting.
- **Suhas, D. L.** and Boos, W. R. (2023). “Understanding projected changes in the Southern African monsoon in CMIP6 models.” AMS 103rd annual meeting.
- **Suhas, D. L.** and Boos, W. R. (2023). “Monsoon depression amplification by horizontal shear and humidity gradients.” AMS 103rd annual meeting.

Journal Reviewer

- Nature Communications, Journal of the Atmospheric Sciences, Journal of Climate, Quarterly Journal of the Royal Meteorological Society, Atmospheric Research, Monthly Weather Review, and Weather and Climate Dynamics.

Teaching Experience

- Teaching Assistant, Geophysical Fluid Dynamics, Indian Institute of Science.
- Teaching Assistant, Atmospheric Dynamics, Indian Institute of Science.
- Teaching Assistant, Probability and Statistics, Indian Institute of Science.
- Science Outreach: Led workshops and educational sessions on science and its applications for elementary and high school students.

Computational Experience

- High-Performance Computing: Experienced in running general circulation models, including CESM and WRF, on high-performance computing clusters.
- Data Handling and Analysis: Skilled in managing, analyzing, and visualizing large datasets, including outputs from high-resolution models and large ensembles.
- Programming Languages: Proficient in Python, Matlab, and R, with working knowledge of C and FORTRAN.
- Machine Learning: Applied TensorFlow, Keras, PyTorch, Scikit-learn, and PyMC3 for machine learning and probabilistic modeling.
- Developed pySWE, a Python solver for the moist shallow water equations on a spherical domain.

Honors and Awards

- Senior Research Fellowship, Indian Institute of Science.
- Grantham Fellowship by Divecha Centre for Climate Change.
- Nikhil Memorial Medal awarded for the best student in Master’s program.